

Assignment

Module 1.5 Assemble and disassemble PC

1. List all the main components you find inside a desktop PC after opening the cabinet and write a short description (1-2 lines) for each.



Main Components Inside a Desktop PC:

Motherboard – The motherboard is the central board that connects all computer hardware parts together, enabling them to function as one unified system.

CPU (Processor) – The CPU handles instruction processing and math calculations. It manages the majority of what your computer does.

RAM – RAM holds temporary data and programs you're actively using. It lets your computer retrieve information fast.

Power Supply Unit (PSU) – The PSU delivers electrical power to all internal parts. It takes electricity from the wall outlet and changes it into the right power for your PC.

Hard Drive/SSD – These store your operating system, software, files, and documents permanently. SSDs typically run faster than traditional hard drives.

Graphics Card (GPU) – The graphics card processes images, videos, and visual effects. It's crucial for gaming and heavy graphics work.

CPU Cooler – The CPU cooler pulls heat away from the processor. It stops the CPU from overheating while it runs.

Cooling Fans – Fans move air through the case and help cool down all the internal parts.

Optical Drive – An optical drive reads and writes CDs and DVDs. Most new desktop computers don't include this anymore.

Expansion Cards – You can plug these cards into the motherboard to add features like wireless internet, better sound, or extra graphics power.

CMOS Battery – This tiny battery keeps your computer's date and time correct. It also saves certain BIOS/UEFI settings when you turn off the PC.

Cables and Connectors – Cables link all the parts inside the case. They send power and move data between the different hardware pieces.

2. **Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC and organize them in a document with captions explaining what you did at each step.**
**

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.**
- I don't have a desktop pc to disassemble, so I am attaching a youtube video that shows all the steps of disassembling of PC and along with that I write all the steps that explain each step of it.
YouTube link: <https://youtu.be/IpFn48RLcNM?si=nEUAYt-S8ukg27Wb>

Desktop PC Disassembly:

Step 1: Switch Off and Disconnect the PC

The computer was turned off completely and unplugged from the electrical outlet before beginning the disassembly.

Step 2: Take Off the Side Panel

The screws securing the side panel were removed, and the panel was carefully taken off to reach the internal hardware.

Step 3: Unplug the Cables

All power and data cables connected to the motherboard, storage devices, and other hardware were disconnected carefully.

Step 4: Remove the RAM

The locking clips on the RAM slots were opened, and the memory modules were gently taken out by holding their edges.

Step 5: Take Out the HDD/SSD

The storage drive was disconnected from its cables and removed from its bracket or slot. It was placed safely to prevent damage.

Step 6: Remove the Graphics Card

The graphics card was disconnected from its power supply, released from the PCIe slot, and carefully removed from the cabinet.

Step 7: Remove the CPU Cooler and CPU

The cooling fan and heat sink were detached from the processor. The CPU was then carefully removed from its socket and stored safely.

Step 8: Remove the Power Supply Unit

After making sure all PSU cables were disconnected, the screws holding the power supply were removed and the unit was taken out.

Step 9: Take Out the Motherboard

The remaining motherboard connections were checked and disconnected. The mounting screws were then removed, allowing the motherboard to be carefully lifted out.

Step 10: Arrange the Removed Parts

All components, screws, and cables were sorted and stored properly. This helps prevent losing parts and makes the reassembly process easier.

3. Reassemble the PC you disassemble, documenting each step with a brief explanation of what you connected or installed and why the order matters.



Reassembly of a Desktop PC:

Set Up the Cabinet – Place the computer case on a clean, level surface and make sure it is disconnected from the power supply. This ensures safe handling of the components.

Install the Power Supply – Put the PSU into its slot and fasten it securely with screws. Leave the power cables accessible for later connections.

Mount the Motherboard – Carefully place the motherboard on the standoffs and tighten the screws. It provides the base for connecting most of the other components.

Insert the CPU – Place the processor into the motherboard socket according to the correct alignment and secure it. The processor needs to be installed before the cooling system.

Fit the CPU Cooler – Attach the heat sink and fan over the CPU and connect the fan wire to the motherboard. This prevents the processor from getting too hot.

Install the RAM – Insert the RAM sticks into the recommended slots and press them firmly until the clips lock them in place. Correctly installed RAM is necessary for the computer to start properly.

Add the Storage Drive – Place the SSD or hard disk in its mounting area and connect its power and data cables. This component stores the operating system and user information.

Fit the Graphics Card – If the computer uses a dedicated graphics card, insert it into the correct PCIe slot and secure it to the case. Attach an additional power cable if the card requires one.

Attach Internal Power Connections – Connect the PSU cables to the motherboard and other components that require power. Make sure each connector is properly seated.

Connect the Cabinet Wires – Attach the front-panel connectors for the power button, reset button, USB ports, and audio to their correct motherboard headers.

Review All Connections – Carefully inspect the inside of the cabinet to ensure that every component is secure and all cables are correctly connected.

Close and Start the Computer – Replace the cabinet panel, connect the monitor and other devices, and switch on the PC. Confirm that it boots normally and that the hardware is functioning.

Why the Sequence Is Important:

The order helps prevent damage and makes installation easier. Components that require direct access to the motherboard are fitted first, while wiring and final checks are completed afterward.

4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process and explain why each is important.



Safety Precautions:

Turn off and unplug the computer – Disconnect the PC from the power supply before opening it to prevent electric shock and protect the components.

Use a clean and dry workspace – Working in a tidy area helps prevent dust, moisture, and other objects from affecting the computer parts.

Prevent static electricity – Wear an anti-static wrist strap or touch a grounded metal surface before handling components to avoid electrostatic damage.

Handle parts with care – Hold components such as the RAM, motherboard, and graphics card by their edges to avoid damaging delicate parts.

Keep screws safely stored – Place screws in a small container so they are not lost or left inside the cabinet.

Avoid applying excessive force – Check out the correct position of a component before installing it. Forcing it may damage the component or its slot.

Keep liquids away from the PC – Liquids can cause short circuits and may permanently damage electronic components.

Record cable positions – Take photos or note where cables are connected before removing them. This makes reassembly easier and prevents incorrect connections.

Tools Required:

screwdriver – Mainly used for removing and tightening screws inside the computer cabinet.

Flat-head screwdriver – Useful for handling certain screws, clips, or fittings.

Anti-static wrist strap – Protects electronic components from electrostatic discharge.

Small container – Keeps screws and other small parts together during the process.

Cable ties or Velcro straps – Help arrange and secure cables neatly inside the cabinet.

Compressed air – Used to blow dust away from fans, vents, and other internal components.

Soft brush or microfiber cloth – Helps clean dust gently without damaging the hardware.

Why These Are Important:

Using proper safety measures and tools makes PC assembly and disassembly **safe, organized, and efficient**. They help prevent electrical accidents, static damage, broken components, and lost parts.